Roll No. ....

TPE–712

B. TECH. (PETROLEUM ENGINEERING)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

RESERVOIR MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain the various technologies for monitoring the reservoir. What are the different technologies for improving the well performance ? Explain in detail. (CO1)

OR

(b) How to build effective team for reservoir management ? What are the various requirement to be a good reservoir manager ? (CO1)

2. (a) What are the five different phases of oil and gas field ? Explain in detail. Explain Lifecycle of oil and Gas reservoir by diagram. (CO2)

OR

(b) Explain field development plan. What is the objective of FDP ? How is it prepared ? (CO2)

P. T. O.

3. (a) What are the factors effecting the effective porosity and permeability ? Define cubical packing, orthorhombic-packing and rhombohedra-packing of spheres form porosity point of view. How much will be the porosity contribution of these packings ? (CO2)

OR

(b) What are the various logging tools for total porosity, primary porosity estimation in the borehole ? What is the tool physics of all tools behind the porosity estimation ? (CO2)

4. (a) Explain primary recovery methods in details. What are the essential factors for this type of recovery ? (CO1, CO2)

OR

(b) When will you apply secondary recovery methods ? What is the factor for achieving optimal production from these methods ? (CO1, CO2)

5. (a) Explain tertiary recovery methods. When these methods will be applied ? What is the essential factor for this recovery methods ? (CO3)

OR

(b) What are the different reserve categorizations ? Explain the different methods for reserve estimation in detail. (CO3)

Roll No. ....

TPE–717

B. TECH. (PETROLEUM ENGINEERING)

(SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ADVANCED EOR TECHNIQUE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Why do we need EOR ? Currently available EOR is based on which parameter ? Explain the different oil production phase. (CO1)

OR

(b) Discuss the methods of EOR. Explain *one* method with schematic diagram for EOR. (CO1)

2. (a) What do you mean by MEOR ? Draw the Schematic diagram of MEOR and explain the each step in increases the production rate. (CO2)

OR

(b) Write and define the different types oil field pressure calculation requirements. (CO2)

P. T. O.

3. (a) Define pressure gradient in the well. Also explain the mathematical expression for hydrocarbon interface (Hc) of gas-oil (GOC), oil-water (OWC). Specify the symbol used. (CO2)

OR

(b) The data of depth and pressure is given in the blow :

Depth (m)	Press. (atm)
0	51.6
200	55.2
400	59.2
600	63.8
800	81.4
1000	99.0
1200	119.0
1400	140.0

Calculate the following :

(i) Pressure gradients

(ii) Depth of gas-oil and gas water interface

(iii) Pressures at GOC and WOC

Given : Depth of top perforation = 1150 m, GOC = 740 m, WOC = 1170 m (CO2)

4. (a) What are the factors on which the decline curves are characterized ? Write down the first condition that must be considered in production-decline-curve analysis. (CO1/CO2)

OR

(b) Discuss the types of rate decline behaviour in the estimation reserves. Write with mathematical expression given by ARPs equation. (CO1/CO2)

5. (a) The following production data are available from a dry gas field : (CO2)

q_t (MMscf/D)	Gp (MMscf)	q_t (MMscf/D)	Gp (MMscf)
320	16000	208	304000
336	32000	197	352000
304	48000	184	368000
309	96000	176	384000
272	160000	184	400000

Estimate the cumulative gas production when the gas flow rate reaches 80 MMscf/D.

OR

(b) Discuss the different selection patterns of water flooding. Discuss the following in water flooding during EOR :

(i) Overall recovery efficiency

(ii) Displacement efficiency

(iii) Volumetric sweep efficiency. (CO2)

Roll No. ....

TPE–719

B. TECH. (PETROLEUM ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
CITY GAS DISTRIBUTION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Using Panhandle B equation calculate the outlet pressure in a natural gas pipeline, with diameter 17.50 in. wall thickness 20 miles long. The gas flow rate is 145 MMSCFD at a flowing temperature of 65OF. The inlet pressure is 1000 psig and the gas gravity and viscosity are 0.6 and 0.000008 lb/ft-sec, respectively. Assume base pressure = 14.7 psia and base temperature = 70OF. Assume that the compressibility factor Z = 0.75 throughout and the pipeline efficiency is 0.95. (Show each step while answering). (CO1)

OR

(b) Using Weymouth equation calculate base temperature for the following data given : Pb = 14.7 psia, D = 11 in, Pipe Length = .0796 mi, gas gravity =.71, T = 75°F, z = .919, p_1 = 570 psia, p_2 = 510 psia, q_h = 272 MMSCFD. (Show each step while answering). (CO1)

P. T. O.

2. (a) For the following data given for a horizontal pipeline, predict Base Pressure : D = 12.09 in, L = 200 mi, e = 0.0006 in, T = 80 °F, Y_g = 0.70, T_b = 520 °R, p_1 = 600 psia, p_2 = 200 psia, z = .9188, q_h = 1076035 cfh. (Show each step while answering). (CO1)

OR

(b) A section of a pipeline system is to handle 200 MMSCFD gas flow. The pipeline inlet pressure is 900 psig and the gas flows to a dehydrator operating at 800 psig. The allowable working pressure of the pipeline is 1480 psi. The following data are given : STC 14.73 psi, 60°F average temperature 80°F, compressibility 0.67, specific gravity of gas 0.85, viscosity 0.01 cp, length 5 miles. Determine D. Using Weymouth equation (Show each step while answering). (CO1)

3. (a) For the following given data for a horizontal pipeline calculate/predict gas flow rate in ft^3/hr through the pipeline : (CO1)

D = 12.09 in

L = 200 mi

e = 0.0006 in

T = 80°F

Y_g = .70

T_b = 520°R

p_b = 14.7 psia

p_1 = 600 psia

p_2 = 300 psia

OR

(b) Using Weymouth equation calculate the natural gas flow rate for gas and pipeline with the given information : $T_b = 60°F$, $P_b = 14.7$ psia, D = 12 in Pipe Length = 500 ft, gas gravity = .65, T = 80°F, $z = .919$, $p_1 = 510$ psia, $p_2 = 490$ psia. (Show each step while answering). (CO1)

4. (a) With the help of Typical Composition of Natural Gas in percentage table describe natural gas composition with compound having minimum and maximum Percentage and also explain natural gas properties. (CO2)

OR

(b) Explain District Regulating Station. Write At least *five-five* advantages and disadvantages of CNG, PNG and LPG. (CO2)

5. (a) Write a short note on city gas station and describe all city gas station operations. (CO1, CO2)

OR

(b) What do you understand by PNG ? Differentiate between Industrial and Commercial PNG and write disadvantages of PNG. (CO1, CO2)

Roll No. ....

TPE–724

B. TECH. (PETROLEUM ENGINEERING)
(SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

PETROLEUM ECONOMICS, RISK AND UNCERTAINTY ANALYSIS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is the significance of petroleum economics in oil and gas industry ? (CO1)

OR

(b) What is the structure of oil and gas industry ? Understanding the structure of oil and gas industry is important for economic analysis. Comment. (CO1)

2. (a) What are stages in oil and gas industry projects ? Explain the significance of each stage. (CO1)

OR

(b) What is the significance of exploration licence in petroleum economics, risk and uncertainity analysis ? Discuss. (CO1)

P. T. O.

3. (a) What is Farm-in and Farm-out agreement in oil and gas industry ? Explain. (CO1)

OR

(b) What is infill drilling ? Explain the significance of infill drilling in petroleum economic, risk and uncertainity analysis. (CO1)

4. (a) What is cash flow analysis in petroleum industry ? Explain by giving examples. What are the objective of cash flow statement ? (CO2)

OR

(b) Explain the direct method of cash flow analysis. (CO2)

5. (a) What is CAPEX ? Discuss the importance of CAPEX in a petroleum project. Explain by citing some example. (CO2)

OR

(b) What is the sunk costs of a petroleum project ? Differentiate between shutdown costs and abandonment costs of a project in oil and gas industry. (CO2)